



ARI BROWN

RESEARCH ENGINEER

PROFESSIONAL SUMMARY

Research Engineer with experience in mechanical systems, materials engineering, experimental design, and data analysis. Proficient in CAD, FEA simulation, materials testing, statistical methods, and Python-based machine learning workflows for engineering applications. Holds a Master of Science degree in Materials Science and Engineering from Louisiana State University.

SKILLS

- Mechanical Systems and Test Engineering:
 - Experimental Design (DOE)
 - Test Method Development & Validation
 - Failure Analysis
 - Materials Characterization
 - Mechanical Testing
 - Metrology
- Materials Science and Fabrication:
 - SEM, XRD, AFM Characterization
 - Hydrophobic Coatings
 - CVD, RIE, Nanoimprint Lithography
 - Additive Manufacturing (FDM 3D Printing)
 - Prototype Fabrication
- Programming and Software:
 - Python, C, C#, MATLAB, SQLite
 - Git and GitHub
 - Jupyter Notebooks

EXPERIENCE

RESEARCH ENGINEER III

ConMed Advanced Surgical | Nov 2021-Oct 2025

- Research & Technology Development:
 - Led the planning, design, and execution of complex engineering experiments (including hydrophobic coatings for electrodes) and research investigations to evaluate new technologies, materials, and product concepts for the Advanced Surgical portfolio.
 - Independently designed, prototyped, and refined innovative solutions to difficult clinical and technical problems.
 - Utilized advanced CAD modeling (e.g., SolidWorks) and analytical tools to inform design decisions.
- Design Control & Documentation:
 - Translated Voice of the Customer and user needs into well-documented design inputs using a Python based GUI I developed for analyzing open ended survey responses using topic models.
 - Led and owned the creation of engineering documentation, including research reports, test protocols, and various elements of the Design History File (DHF), in accordance with the Quality Management System (QMS).
- Verification & Validation (V&V):
 - Assisted with the development and implementation of research and test methods for product V&V activities.
 - Executed preclinical and tissue testing to ensure new designs meet safety, efficacy, and compliance standards.
 - Performed comprehensive data reduction and statistical analysis using engineering tools to provide insightful results and technical feedback.
- Cross-Functional Collaboration:
 - Effectively worked independently with minimal supervision in a cross-functional team environment (R&D, Marketing, Quality, Manufacturing, Regulatory) to drive Phase 0 Projects.

MECHANICAL DESIGN ENGINEER AND R&D SPECIALIST

Benoit Premium Threading | Nov 2019-Nov 2021

- Research & Technology Development:
 - Designed and conducted experiments such as weight loss corrosion studies of mild to low carbon steels
 - Investigated and systematically tested the feasibility of new coating technologies for oil and gas tubular goods, with the objectives of reducing friction, increasing corrosion resistance, and increasing abrasion resistance.
 - Leveraged FEA software to aid in prediction of material failure during plastic deformation
 - Interpreted mechanical testing data and material composition to identify appropriate parameters for material recovery after plastic deformation
 - Programmed in Python C++ for data analysis and automation of task along with utilization of Pandas and SQLite for data science.
- Cross-functional Teamwork:
 - Assisted in customer questions relating to material sciences
 - Review and approve colleague CAD drawings
- Mentorship and Community Outreach:
 - Provided technical guidance for Mechanical Engineering senior design project funded by our company.
 - 3D printed over 500 ear savers and made 100 face shields for local hospitals during COVID 19 pandemic.

SKILLS (CONT.)

- Design and Simulation:
 - SolidWorks and AutoCAD
 - Finite Element Analysis:
 - COMSOL Multiphysics and ANSYS
- Machine Learning & Data Analytics
 - Anomaly Detection
 - Time-Series Sensor Analytics
 - Feature Engineering (Rolling Statistics, Signal Velocity, FFT/Spectral Features, ROC AUC feature ranking)
 - Isolation Forest and Random Forest.
 - Model Validation (Walk-Forward Validation, Precision-Recall Analysis, AUC/F1 Evaluation)
 - Pandas, NumPy, Scikit-learn
 - Statistical Analysis (DOE, ANOVA, Regression)
 - Minitab
 - Topic Modeling: BERTopic

EDUCATION

MASTERS OF SCIENCE, MATERIALS SCIENCE AND ENGINEERING LOUISIANA STATE UNIVERSITY | 2018

GPA 3.333

- **Teaching Assistant:** Instructed undergraduate students on material behavior, including x-ray diffraction, heat treatment processes, aluminum alloy hardening, hardness and impact testing, tensile testing of metals and polymers, and scanning electron microscope (SEM) concepts.
- **Research Assistant:** Used finite element analysis (COMSOL) for Level Set modeling of pool boiling. I also fabricated super-hydrophobic substrates for condensation experiments using thermal imprinting and self-assembly monolayers (SAM).
- **Leadership & Outreach:** Guided a high school student in developing research skills, focusing on creating and testing super-hydrophobic surfaces. Further assisted in a STEM outreach program, introducing elementary students to nanotechnology through hands-on experiments with water-repelling surfaces and molding techniques.

BACHELORS OF SCIENCE, MECHANICAL ENGINEERING SOUTHEASTERN LOUISIANA UNIVERSITY | 2014

GPA: 3.420

- **Activities and societies:** Honor's Society
- Completed an undergraduate thesis using COMSOL (FEA software) to model stress concentrations at various angles of attack, identifying the angle that produced the highest stress.

PROJECTS

Industrial Predictive Maintenance Anomaly Detection Framework Independent Research Project (July 2026-Present)

Developed a Python-based unsupervised machine learning pipeline to detect industrial equipment failures from multivariate sensor time-series data. Engineered temporal, mechanical, and frequency-domain features and evaluated Isolation Forest anomaly detection performance using chronological walk-forward validation.

- Current Status: Refracturing and restructuring

Surgical Smoke Removal in Laparoscopic Video: A Domain Adaptation Study Independent Research Project (June-August 2026)

- Investigated the transferability of AOD-Net, a CNN originally designed for outdoor atmospheric haze, to surgical smoke removal on real paired laparoscopic footage (DesmokeData, MICCAI 2024)
- Designed and implemented a full evaluation protocol (PSNR/SSIM on held-out validation data) benchmarked against published dehazing baselines
- Conducted a density-stratified analysis revealing a systematic performance gap (~6.5 dB PSNR) between light and dense smoke conditions, identifying smoke density as a key confound not addressed by the base architecture
- Findings suggest architecture capacity and density-aware training strategies as concrete directions for improving robustness in surgical desmoking.